2024 IEEE International Students' Conference on Electrical, Electronics and Computer Science (SCEECS) | 979-8-3503-4846-0/24/$31.00 ©2024 IEEE | DOI: 10.1109/SCEECS61402.2024.10481917

Digital Image Forgery Detection using Convolutional Neural Network (CNN): A Survey

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***Abstract*—Nowadays, there are lots of pictures everywhere because almost everyone has a camera or a smartphone. This has led to a worrying trend: people change images for different reasons. While digital images are powerful tools for information exchange, they are also susceptible to exploitation for political, cultural, economic, and social purposes through the dissemination of falsified information. The widespread availability of sophisticated image editing software has significantly lowered the barriers to image manipulation, making it challenging to detect such forgeries. These manipulated images can have far-reaching consequences, affecting fields such as medical diagnosis, legal judgments, politics, and insurance claims. This paper delves into the realm of forgery detection, a vital domain within machine vision. It comprehensively examines the techniques employed to manipulate digital images and offers a detailed review of existing methodologies for digital image forgery detection.**

Keywords—Digital Image Forgery Detection, CNN, Deep Learning, VGGNet, ResNet

1. INTRODUCTION

In current culture, digital photos are used to communicate. With the development of cameras and cell phones, many images are taken daily. In 2020 alone, 1.4 trillion digital photos were created [1]. These photos, which have many uses and a big impact, are quickly shared worldwide on Instagram, Telegram, Facebook, and the Web. Image use is rising due to its simplicity and efficiency in delivering thoughts and information compared to written sources. However, the ease of access and speed of image sharing have created a disturbing tendency. Many people and groups manipulate visual images to spread false information for political, cultural, economic, and social gain [2].

The proliferation of sophisticated picture editing software, such as Adobe Photoshop, which is offered in both private and open-source iterations, has facilitated the process of image alteration. Changing digital photos without showing any signs is really hard. Using changed pictures in a bad way can cause a lot of problems in areas like medical checks, court decisions, arguing about patents, talking about politics, and making insurance claims. The process of image manipulation encompasses the concealment or addition of inaccurate information to digital photographs, so undermining its fundamental attributes such as structural integrity, texture, color, and frequency, ultimately rendering them invalid [3]. Unethical editing of medical CT scan pictures shows those without COVID-19 as ill [4]. In another incident, a journalist's shot from the first day of the 2017 G-20 conference in Germany was altered to include the Russian president. The edited image was widely shared on social media and news sources, causing confusion and debate. Picture fraud can affect political decision-making, movements, and international conflicts. Development and implementation of forgery detection methods are urgent [3]. This study examines counterfeit detection, a vital machine vision field. This study extensively examines digital photo manipulation methods and digital image fraud detection methods. Figure 1 shows the active/non-blind and passive/blind approaches, together with their subcategories [4-6].

This work aims to provide insights and methods for addressing the growing concerns of digital picture manipulation in our connected global environment. The active approach entails actively participating in the topic or work. The individual must take the active technique involves pre-processing the image to add information. Common active methods include digital watermarking and digital signatures. This method involves professional software and technology and the source image to integrate or extract information. Without these criteria, the active approach is impracticable and inefficient. Given the criteria, the active approach makes forgery discovery rather easy [7]. The passive, or blind, method does not require image pre-processing. This method's main goal is to identify image changes and discrepancies by examining its content and structure. Since it requires no prior knowledge, the passive method is popular. This includes copy-move fraud detection, image splicing analysis, image retouching detection, and object removal recognition [8].

Passive Approaches Categories

Survey breadth and focus relate to inquiry and concentration areas. This broad survey covers passive forgery detection methods. These categories include:

The term "copy-move" (CM) refers to copying a section. Create or change a document, item, or signature to forge it. The widely used and simple image-altering technique involves replicating and inserting pieces of an image into different areas of the same image. The main goal is often to hide important facts or include false stuff. Since they came from the same image, the repeated segments integrate structurally and texturally. Images are often rotated, scaled, and post-processed using blurring, brightness changes, compression, and noise. The above changes make them hard to spot [9, 10].

Image splicing (SP) combines multiple photos into one. Splicing creates boundaries and areas of decreasing sharpness. Images can be edited to seamlessly contain these areas, making them invisible to the human eye [4,11].

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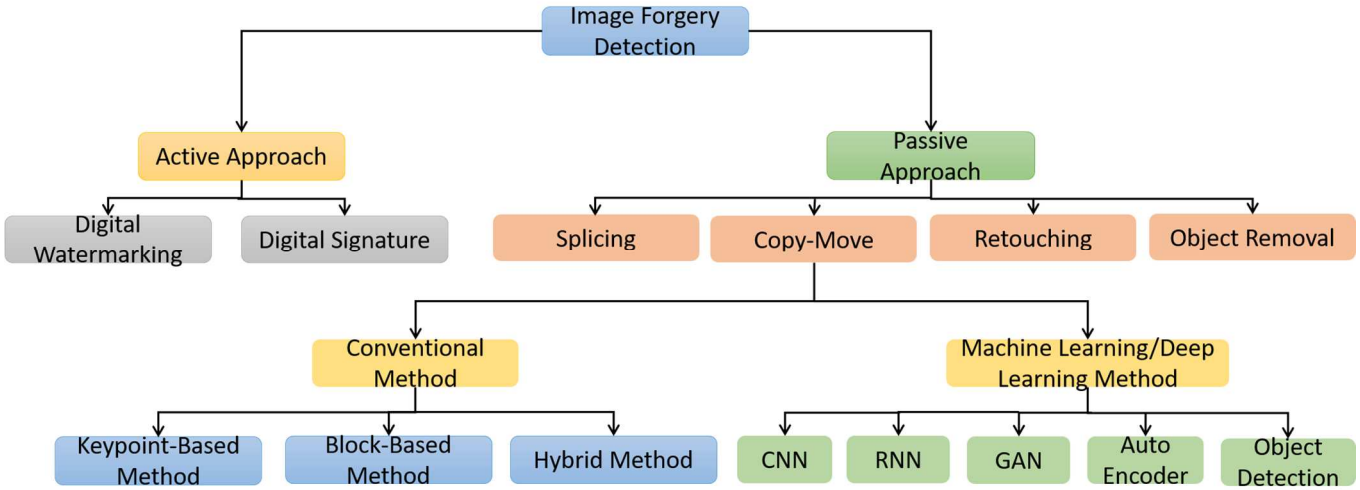

Figure 1: Image forgery detection categorization.

Compared to other image production methods, retouching is less severe. Pixel characteristics are amplified or reduced [12]. Photo editing software and publications use the above method to increase or decrease pixel hue.
Object removal, a more intrusive forgery, can change an image's meaning. Copy-move and image inpainting are the basic object removal methods. Copy-move involves overlaying a section of a picture from the same or a different image over an object to be removed. This feature is liked for its simplicity. Picture inpainting (IP) was invented to repair damaged photos and remove scratches from old ones. It does this by eliminating elements from an image and replacing them with nearby pixels, ensuring textural and structural consistency [13].

The purpose of this survey is to methodically collect and evaluate data and information in order to gain a comprehensive understanding of a certain topic or phenomenon. In recent times, there has been a growing focus and significance attributed to the field of forgery detection. Consequently, extensive research has been conducted on conventional and deep learning approaches. In light of extensive research [2, 14], the present study undertakes a comparative analysis of forgery detection systems, aiming to elucidate their respective strengths, limitations, and associated difficulties.

This work investigates the application of deep learning algorithms in forgery detection from two novel angles, considering the significant achievements shown in diverse problem domains with the use of appropriate hardware and datasets. The focus of this study is the application of deep learning techniques to detect counterfeiting. The text emphasizes several approaches and structures that are special to deep learning. The report additionally examines benchmark datasets for copy-move, splicing, and inpainting forgery analyses.

II. Forgery Detection using Deep Learning (DL)

In recent years, there has been a notable growth in the utilization of deep learning algorithms for counterfeit detection, with a special emphasis on their application in the field of computer vision. Deep learning can autonomously extract hierarchical features from data, hence streamlining the process of creating semantic representations without the need for manual feature engineering. One limitation of deep learning is the substantial amount of training data that is necessary [15]. Numerous potential resolutions have been suggested in order to effectively tackle this matter.

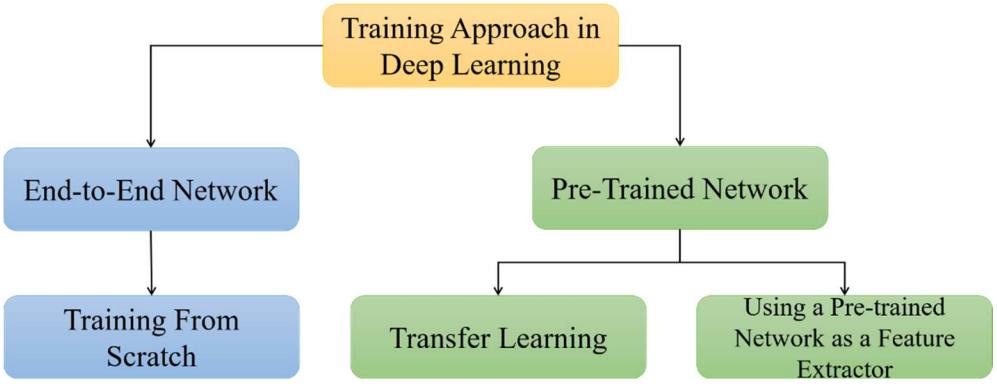

Figure 2: Deep learning Training approaches

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Table 1: Different Methods for Feature Extraction in Image forgery techiniques

Reference	Forgery type	Summary	Dataset	Outcomes
[26]	CM	(The utilization of pre-trained convolutional neural network models, such as SqueezeNet, MobileNet v2, and ShuffleNet, in conjunction with a support vector machine (SVM) classifier, is a common approach in academic research.	MICC-F220	This approach involves employing decision fusion, which enhances the effectiveness of the proposed network.
[24]	SP	The proposed methodology involves the combination of three distinct components: the Discrete Wavelet Transform (DWT), the Local Binary Patterns (LBP) algorithm, and the Residual Network (ResNet). Additionally,	CASIA	This technique combines both manual features and deep learning, resulting in a substantial number of feature dimensions.
[25]	CM, SP	The utilization of the Inception v3 model in conjunction with the Nearest Neighbor algorithm.	,CASIA V2.0,CoMoFod	The observed phenomenon exhibits a notable degree of resistance when subjected to post-processing processes.
[20]	CM	The utilization of trigonometric transformation, discrete wavelet transform (DWT), and convolutional neural networks (CNN) is being explored in academic research.	MICC-F220	This methodology employs a combination of block-based and deep-learning methodologies. Additionally, the system that has been developed exhibits adaptability in its ability to detect forgeries using an active approach.
[22]	CM	The integration of clustering techniques with the VGGNet model. The matching algorithm is a computational procedure designed to pair or align entities based on specific criteria or similarity measures. It is commonly employed in	MICC-F220	It maintains its robustness even when subjected to geometric transformations and the introduction of noise
[23]	SP	The Resnet+ network architecture. The classification algorithms K-Nearest Neighbors (KNN), Support Vector Machines (SVM), and Naive Bayes are	CASIA V2.0	This approach capitalizes on the benefit of employing decision fusion.
[26]	CM	The proposed approach involves pre-training the AlexNet architecture followed by the utilization of a Support Vector Machine (SVM) for further processing	MICC-F220	The object has a high degree of robustness when subjected to various geometric modifications.
[27]	CM	The utilization of a pre-trained AlexNet model in conjunction with logistic regression.	MICCF60,MICCF2000	This approach employed the pre-trained AlexNet model for feature extraction.
[32]	SP	The utilization of transfer learning methodology in conjunction with pre-trained EfficientNetB model.	CASIA	The EfficientNetB network outperforms other deep learning architectures, especially when considering factors like increased network depth, neuron count in layers, and enhanced input resolution.
[30]	CM	The proposed approach involves the utilization of a dual branch CNN architecture, whereby each branch is equipped with distinct filters and kernel sizes.	MICCF2000	The functional API method in this technique utilizes a dual branch of convolutional neural networks (CNNs) to boost robustness against scaling. This is achieved by incorporating filters with varied kernel sizes in the two branches.
[28]	CM, SP	The utilization of transfer learning is explored in this study, specifically focusing on pre-trained VGG16, VGG19, ResNet50, and DenseNet networks.	CASIA V2	The methodology employed in this study involved utilizing pre-trained neural networks and implementing transfer learning techniques. Furthermore, ELA (Error Level Analysis) photographs were employed as input to accentuate the areas of falsification.
[31]	CM, SP	This study explores the application of CNN in the detection of image forgery. Specifically, it investigates the effectiveness of	CASIA	This approach incorporated a hybrid of deep learning and block-based methods. Furthermore, it not only identifies forged images but also distinguishes the type of forgery, such as Copy-Move and Splicing.

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		CNN in identifying Copy-Move (CM) and Splicing (SP) forgeries. Additionally, the study examines the utilization of a Block-Based approach with Zernike Moment for the detection of copy and pasted regions within images.		
[29]	CM	The utilization of a pre-trained AlexNet model in conjunction with a block-based technique.	GRIP	This methodology integrates a fusion of deep learning characteristics derived from a pre-trained AlexNet model alongside block-based techniques.
[19]	IP	The utilization of pre-trained ResNet models	The procedure entails the generation of an inpainting image utilizing the Imagenet dataset..	The proposed methodology employs high-pass filtering of images in order to enhance the forged region.

Deep Learning Training Methods

Two common training methods for deep learning network forgery detection have been found:

Pre-training-free training

This method creates a network from scratch. Network training involves a lot of data and time. This technique can alter architecture and settings, but it is less practical and successful in scenarios with little datasets [16-17].

Transfer Learning

Transfer learning uses AlexNet, VGGNet, GoogLeNet, and ResNet, which have been trained on ImageNet. Generalization is high in these models. Transfer learning uses a pre-trained network's beginning and intermediate layers for edge and form detection and fine-tunes the final layers to fit the task and dataset [18-19].

III. Feature Extraction using Pre-trained Models

Existing models can extract important features from new inputs, making them excellent feature extractors. The feature maps discussed above have the potential to be directly used in classification tasks or they can be employed as feature vectors in the field of machine learning. This methodology utilizes the natural attributes contained within the basic layers of the current network, hence obviating the necessity for retraining the model. Feature extraction entails the application of pre-existing models such as AlexNet, VGGNet, ResNet, and Inception v3.0 for the purpose of extracting relevant features. Feature vectors are determined by the utilization of various machine learning algorithms, such as Support Vector Machines (SVM), K-Nearest Neighbors (KNN), Decision Trees, Naive Bayes, or shallow neural networks. Previous research has demonstrated that decision fusion can improve categorization in specific instances [21], [23], [24]. The detection of counterfeiting has been facilitated by researchers through the utilization of both manual methods and deep learning techniques. The approach described in this study integrates traditional techniques with deep learning methodologies in order to enhance the accuracy of detection [24]. In summary, the utilization of deep learning-based forgery detection encompasses a diverse range of training and feature extraction techniques, each possessing distinct merits and suitability for various datasets and scenarios.

Image Forgery Detection with the help of Deep Learning.

This section provides an overview of different deep learning architectures employed in the field of forgery detection. CNNs, Recurrent Neural Networks (RNNs), and other associated architectural configurations have gained significant popularity in the field of deep learning. These architectural models have the capability to identify various forms of photo counterfeiting, offering several advantages in the process.

Forgery detection using CNN

Due to their architecture for learning spatial hierarchies of data, CNNs are becoming more popular for forgery detection. These networks excel in capturing low-to-high-level visual patterns. Various apps have used these techniques to detect copy-move, splicing, pixel, and image fraud. Pre-trained CNNs like AlexNet, VGG16, and Inception v3 are effective feature extractors for forgery detection. Hybrid approaches that combine deep learning features with block-based and pre-processing methods to improve detection accuracy have been studied extensively. Transfer learning also uses pre-trained models to fine-tune counterfeit detection tasks. Error Level Analysis (ELA) images, which highlight fraud, are combined with CNNs to improve detection. Functional API models can strengthen dual-branch CNNs against scale. The MICC-F2000 and GRIP datasets were used to test these models. Utilising heatmap visualisation methods with Convolutional Neural Networks have been used to detect image and pixel counterfeiting. This method helps identify forged image portions. The efficacy of these methods has been tested on CASIA V2, CASIA, MICC-F600, and MICC [19], [29], [28], [30].

Extra Deep Learning Architectures

EfficientNetB and other deep learning architectures have been studied for forgery detection. The performance of these architectural solutions is better than standard CNNs. The researchers improved counterfeit identification by considering depth, layer number of neurons, and input resolution. These architectures were tested using CASIA [32].

In summary, several deep learning architectures have been used to detect forgery, each with its own advantages and applicability for different detection tasks and datasets. These architectural designs have advanced fraud detection in the field, enhancing precision and resilience.

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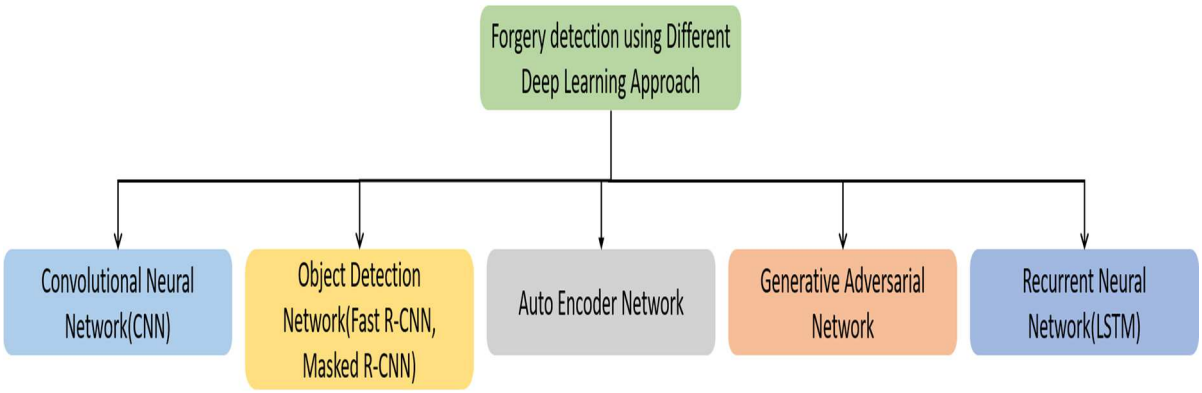

Figure 3: Forgery detection using Deep Learning

Conclusion

This study offers an extensive examination of the detection of digital picture counterfeiting, which is a crucial field within the realm of machine vision. This statement highlights the increasing difficulties associated with the manipulation of digital photographs in the contemporary networked society. It explores different passive methods employed to identify instances of image forgery. Additionally, this paper emphasizes the notable contribution of deep learning in the domain of forgery detection. It also examines several training methodologies and deep learning frameworks employed within this discipline.

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